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# 320413 (20)

**BE (4th Semester)**

**Examination, Nov.-Dec., 2010**

**Branch : Civil**

## SURVEYING - II

***Time Allowed : Three Hours***

***Maximum Marks : 80***

***Minimum Marks : 28***

**lote :** Attempt all questions. All questions carry equal marks. Part 'a' is compulsory and attempt any





two parts from rest of three parts b, c & d in questions.

### UNIT - I

**Q. 1.** (a) Name different types of ED instruments.

(b) What are the different types of triangulati system ? Explain in brief. 7

(c) What is meant by the strength of figure How would you determine it.

(d) What is the extension of the base ? How it done.

## UNIT - II

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2. (a) Define most probable value. 2

(b) What do you understand by propagation of errors ? Give various laws of propagation. 7

(c) Discuss the use of the theory of least squares. How the normal equations are formed in practice ? 7

(d) What do you understand by station adjustments and figure adjustments ? How would you perform station adjustments. 7

## UNIT - III

Q. 3. (a) Define range finders.

(b) Explain how a substense bar is used with theodolite to determine the horizont distance between two points. 7

(c) What are the different methods employe in tacheometric survey ? Describe th method most commonly used.

(d) Two distances of 20 meters and 10 meters were accurately measured out an

the intercepts on the staff between the outer stadia webs were 0.196 m at the former distance and 0.996 at the latter. Calculate the tacheometric constants. 7

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## UNIT - IV

Q. 4. (a) Define phototheodolite. 2

(b) How do you determine the scale of an aerial photograph ? What do you understand by the terms 'datum scale' and average scale ? 7



(c) Describe the various steps involved in the combination of vertical air photographs by the principal point radial line method. 7

(d) Write short notes on : 7

(i) Stereoscopic vision.

(ii) Mirror stereoscope.

(iii) Parallax bars.

## UNIT - V

Q. 5. (a) Define soundings. 2

(b) What do you understand by shore line survey ? Explain in brief. 7

(c) Write short note on equipment needed for soundings : 7

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(i) Sounding boat.

(ii) Sounding rods or poles.

(iii) Lead lines.

(iv) Sounding machine.

(v) Fathometer.

(d) List different types of method of plotting of soundings. Explain in brief any one of them. 7